PipedPeer: Zero-Configuration Distributed Computing via Recursive, Decentralised Coordination

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***Abstract*—Distributed computing frameworks presuppose a configured cluster, a program written against their API, and machines that can reach one another. PipedPeer addresses all three for Python: a single command runs an unmodified script on other machines, each running the same daemon, with no head node and no scheduler process. The script's environment is built as a Nix closure against a pinned package set, so it has the same store path on every machine, and a peer receives only the store paths it lacks. Placement runs in the submitting client under a lease, and every distributed path falls back to a local one. The parallel primitives the script already uses are substituted at import time; parallel maps are divided by a makespan solver over measured peer throughput, and numerical work moves only when a cost model predicts a gain. PyTorch data-parallel training is synchronised through the daemons with batches weighted by measured speed, either exactly or by DiLoCo-style periodic averaging. At the default two-rank interval of 32 steps, averaging stayed within 1% of one device's validation loss on every test model in an in-process simulation (mean of three seeds), and in a separate container-lab run (one seed) each rank sent 7.6–9.2$\times$ fewer bytes than exact synchronisation. Machines behind different routers meet through an introducer that any member can serve, then connect directly over QUIC, authenticated by node keys and a cluster secret; no data is relayed; NAT traversal is evaluated on a two-site container testbed. A phone joins as a chunk worker. In a container lab, a submission whose closure the worker held took a median of 0.83 s against 23.4 s for the first, and interception cost $1.24\times$ the startup of a trivial job. Between two laptops in different homes over Tailscale, a CPU-bound script took about 14 s against 18 s on one.**

***Index Terms*—distributed computing, peer-to-peer, NAT traversal, transparent interception, reproducible environments, content-addressed caching, cost-based scheduling**

I. Introduction

The gap between running `python script.py` and running the same work on several machines is large in operational effort and often small in the computation involved. Apache Spark [1], Ray [2] and Dask [3] presuppose a configured cluster with a head node, matching software on every worker, a scheduler, and a network on which every machine reaches every other. Remote execution over `ssh` avoids the setup but offers no environment management, placement, isolation or fault tolerance.

Three obstacles are usually treated separately. The *environment* obstacle: a remote machine must be made to match the local one. The *code* obstacle: using more than one machine normally means rewriting the program. The *reachability* obstacle: machines behind different home or campus routers accept no inbound connections, and the usual answers — a VPN, port forwarding or a relay — are either configuration or a bottleneck. PipedPeer addresses all three. Its contributions are:

1) **Peer-to-peer closure distribution.** Building on Nix's content-addressed store paths [4], we push signed closures between peers, send only the store paths a receiver lacks, and offer distributed work only to peers that can run the closure (Section III-B).
2) **Transparent interception with measured work sharing.** Parallel primitives are substituted at import time, parallel maps are divided by a makespan solver over measured rates, and numerical work is gated by a cost model (Sections III-D, III-E). The division is *recursive*: the node that holds a map splits it among its peers, and each peer splits its part again across its own cores.
3) **Joining across networks without a relay**, through an introducer that only exchanges addresses and that every daemon serves (Section III-C).
4) **Placement without a scheduler**: filters and a bounded score evaluated in the submitting client, with a three-phase lease (Section III-A).
5) **A fallback that preserves the result on every distributed path** (Section III-F).

TABLE I
COMPARISON WITH EXISTING SYSTEMS. THE LOWER BLOCK IS WHERE OURS IS PARTIAL OR THE WEAKER SYSTEM

Property	Ours	Spark	Ray	Dask	ssh
Runs unmodified Python	yes	no	part.	part.	yes
No cluster to configure	yes	no	no	no	part.
No head or master node	yes	no	no	no	yes
Reproduces the environment	yes	part.	part.	part.	no
Joins NATed machines, no relay†	yes	no	no	no	no
Survives losing a node*	part.	yes	yes	yes	no
Authenticated, encrypted	part.	yes	part.	part.	yes
Sandboxes each job	part.	no	no	no	no
Runtimes beyond Python	**no**	yes	part.	no	yes
Proven in production	**no**	yes	yes	yes	yes

†When one introducer is reachable; otherwise a public address or an overlay is needed.
*Any worker; not the submitter of a parallel script or the machine of a training ring's rank 0.

6) **Speed-weighted data-parallel training through the daemon**, with exact or fast (DiLoCo-style) synchronisation chosen by ring size (Sections III-D, IV-D).
7) **An unprivileged sandbox and a phone worker**: `crun` with a memory ceiling and GPU driver files bound one by one; Android joins as a chunk worker.

II. Related Work

Spark [1], Ray [2] and Dask [3] are mature and far more capable at scale. Ray ships a drop-in `multiprocessing.Pool` and a `joblib` backend, and Modin [5] provides distributed pandas behind the pandas API; each needs a changed import and a running cluster. PipedPeer changes no line of the program, because substitution happens before the first user statement, and it decides per call whether to distribute. Table I summarises the differences. BOINC [6] pioneered volunteer computing, but as a platform for specific projects, not a general executor; HTCondor [34] scavenges idle workstations through a central manager and job descriptions.

The cross-network mechanism follows established practice. Reflexive addresses are discovered as in STUN [7]; port mappings are requested with PCP [8], NAT-PMP [9] and UPnP-IGD [10]; simultaneous UDP probing from the socket that will carry the connection is the hole punching of Ford et al. [11]; links are QUIC [12].

Edge-computing offloading studies the decision our cost model makes. Deng et al. [13] weigh transmission against local computation; Yong et al. [14] show partial offloading outperforming all-or-nothing, the regime our interception works in; learned policies [15] exceed a fixed rule at the price of training; and topology-aware placement [16] outperforms scalar scoring, a limitation our placement shares. Spenger [17] defines failure transparency, under which application code never observes a failure; Section III-F must provide it for scripts that contain no failure handling. For training we build on PyTorch DDP [18], whose ring all-reduce Horovod [35] popularised and which assumes ranks that reach their neighbours directly, 8-bit gradient compression [19] with error feedback [20], and periodic averaging as in DiLoCo [21] and local SGD [22].

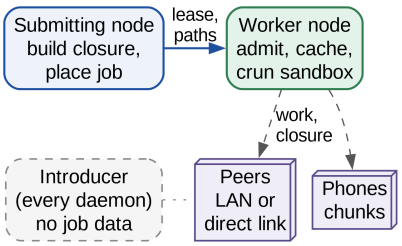


Fig. 1. System design. Both roles are the same binary; peers on other networks are reached over direct links the introducer helps set up but never carries.

III. System Design

Design goals. (G1) The script runs unchanged. (G2) Joining needs one setup command and, across networks, one invite code. (G3) No machine is special: there is no head node, scheduler or registry, and every machine both submits and works. (G4) Distribution never changes the result and is declined when it would be slower. (G5) No job data passes through a third machine.

Coordination is *decentralised*: every daemon serves the introducer for its cluster, placement is computed in the submitting client, and a training ring's lead is the node running its rank 0. A node that cannot hold a map forwards it whole, once, to a peer that splits it; a part is never split across machines again.

Every machine runs a single daemon; the command-line client is short-lived. The coordinator is a library linked into the client, so placement is decided where the job is submitted (Fig. 1). On one network participants find each other unaided; across networks a participant runs `pipedpeer connect` with a one-line code that another member's `pipedpeer invite` printed, carrying the cluster secret and a few introducer addresses.

A. Placement

Candidates come from known peers, LAN discovery and internet-linked peers. Hard filters run in order — architecture, GPU if required, then free memory, cores and VRAM — and the remaining candidates are scored:

$$S = \Big(1 - \frac{u_{\text{cpu}} + u_{\text{mem}}}{200} - 0.05\, n_{\text{jobs}} + 0.10(\hat{c} - 0.5) + 0.06(\hat{f} - 0.5) + 0.10(\hat{m} - 0.5) + 0.10(\hat{r} - 0.5)\Big) \cdot h \tag{1}$$

with utilisations u in percent, n_{jobs} running jobs, capacity terms $\hat{c} = \min(\text{cores}/32, 1)$ and $\hat{f} = \min(\text{MHz}/4000, 1)$, headroom terms $\hat{m} = \min(\text{free GiB}/32, 1)$ and $\hat{r}$ (free fraction of cores), 0.5 for a missing metric, and h the stored health

score (1.0 for a node that answers). S is clamped at zero; a GPU request adds up to 0.8 from VRAM, devices and compute capability. The utilisation terms can subtract 1.0, the four normalised terms only 0.18 either way. Free memory is what a node can *grant*: where a user `systemd` is available, each daemon runs under a cgroup ceiling of half the machine, and co-located daemons (several on one machine) subtract one another's reservations.

Work is reserved by a three-phase lease; the concurrency and memory checks and the insertion happen under one lock. The submitter renews every 30 s, also while the environment builds and uploads, against a 5-minute expiry. A submitter that dies stops renewing and its slot is reclaimed; a worker that dies has its job rescheduled with back-off.

B. Environment Reproduction

The client scans imports, resolves them to packages, and builds a Nix closure from a flake pinned to an *exact* package-set revision, so the store path is a stable name for an environment across machines and time. Imports absent from the mapping table are looked up by name in the pinned set, the scan follows the script's own modules, a `requirements.txt` is honoured, and an import still unresolved is reported before the run.

Before uploading, the client asks whether the target can already run the closure and otherwise sends only the store paths it lacks; the worker then pushes the closure to peers that cannot run it, since only peers that can are offered distributed work. Between daemons the push is a binary cache produced and imported by Nix itself and signed with the sending node's key; the receiver checks that key against the signer's published identity before trusting it for that import. Closures go only to peers of the same architecture; a peer of another architecture receives the package list and builds its own environment once. A phone joins under Termux [23], where `proot-distro` provides, through `proot` [24], a Debian root with single-user Nix, and takes chunks only.

C. Joining Across Networks

An introducer keeps an address book, keyed by a hash of the cluster secret, and forwards punch requests between two nodes; it carries no job data. Every daemon serves one for its own cluster and registers with every introducer it knows, so no single machine's address is a point of failure. Each node offers its LAN addresses, a port mapped on its router (published only when the router's external address matches the one the introducer observed) and its reflexive address. Probes, introducer messages and QUIC share one UDP socket, because a router's mapping belongs to the socket that opened it; both sides probe every candidate pair every 250 ms for 15 s. When one side's NAT allocates a new port per destination, that side opens 256 sockets towards the peer while the other sends 600 probes to random ports, and the socket that is hit is kept [25]: about 91% of rounds over 64,511 ports by calculation, not measured.

The winning path carries QUIC with a hello signed by each node's Ed25519 [26] key and checked against the node it meant to reach; the hello signs the session's exported keying material [27] and proves the cluster secret by an HMAC [28] over it, so it cannot be replayed on another session. The link appears locally as a loopback forwarder recorded as one path to that node, so placement, transfer and gradient exchange run over it unchanged. A node is identified by its key's fingerprint, never by its address: the node store keeps every path to it and switches only to a path that answered and proved the key, and QUIC connections migrate when an address changes, so a machine that changes network rejoins on its own. A peer with no path is reported and routed around, never relayed. The daemon API requires the cluster secret on every route and serves pinned TLS [29] and HTTP on one port.

D. Transparent Interception

A Python package embedded in the binary is placed on the interpreter's path so that it is imported before the user's first line; libraries are patched on first import. Substituted primitives are parallel maps, `joblib`, pandas grouping, joining and chunked reading, NumPy matrix products and decompositions, and data-parallel training.

Parallel maps. A kernel travels as *source* when it is a module-level function, including one in `__main__`, or *by value* through cloudpickle [30] when it is a lambda or closure. The node that holds a map splits it once among its peers, and each peer runs its part on its own cores. The split is a makespan solver: each device is modelled as a setup time plus a rate, fitted from the parts it has returned, and a device whose setup alone exceeds the makespan is declined. A part unfinished past its prediction counts as at least that long, a part waiting or overdue is copied to a free peer, and the first copy to finish is kept.

DataFrame operations. pandas grouping and joining hash rows by key, so each key is complete on a single node and any aggregation is exact with no combiner; the row order of joins is not preserved.

Training. Ranks post gradients to daemons rather than to one another. Each rank posts to the lead node's daemon; in exact mode on three or more directly reachable machines, it posts one shard to each rank's daemon (a butterfly). Every rank receives one sample-weighted average $\sum_i n_i g_i / \sum_i n_i$, sent as 8-bit values with per-block scales and error feedback. The global batch is partitioned across ranks in proportion to their speed, preserving the effective learning rate. Unless the user fixed the ring size with `--ddp` N, after a warm-up the lead compares the split's step time with the fastest rank alone and keeps the ring only if it is faster. Admission takes the k fastest candidates maximising $k\,r_k$, the equal-batch throughput, a conservative bound since placement cannot price synchronisation. Samples go to ranks by a weighted rendezvous hash [31]: when a rank dies the others take its share at the next epoch boundary with over 95% of their samples in place, and a returning rank joins from the lead's snapshot, taking about $1/n$ of them.

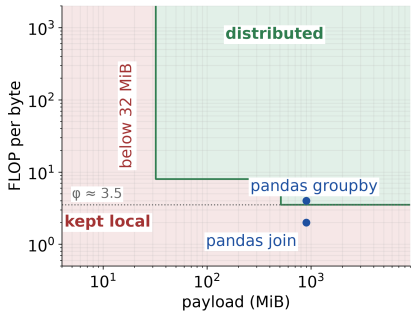


Fig. 2. Decision regions for pandas work ($\rho = 1$, $R = R' = 1$ GFLOP/s) with three nodes on a 500 MB/s link. Above 512 MiB, Equation 2 distributes work above about 3.5 FLOP/B.

A fast mode synchronises every H steps: an exact warm-up (a tenth of the run, 10–500 steps), then local steps whose weights are averaged by samples, with optimiser moments averaged every tenth round; H shrinks to 1 over the run's last tenth. It never changes epochs, batch or learning rate, and must keep validation loss at most 1% above one device's (mean over seeds, every test model). DiLoCo's outer Nesterov step (lr 0.7, momentum 0.9) applies in the steady state $0.7/(1-0.9) = 7$ times each round's averaged change; DiLoCo syncs every few hundred steps, and at our intervals of 2–64 steps the amplified update failed the bar, so PipedPeer uses plain averaging (outer lr 1, no momentum) and chooses H by ring size (Section IV-D). Automatic mode never picks fast sync past four ranks; an explicit fast ring above four runs at $H = 4$ with a warning. Under automatic sync the lead switches a ring to fast mode once, when the predicted fast step with a 10% margin is shorter than the exact split's and, for a ring not fixed by `--ddp` N, also shorter than the fastest rank's alone; under `--ddp` N the choice is between the two sync modes only.

E. The Cost Model

Numerical work is gated. With ϕ operations and ρ link crossings per byte, bandwidth B, local and per-node remote throughput R and R', and K nodes holding the environment (the submitter included), divisible work moves only when its time per byte is lower on the cluster:

$$\frac{\phi}{R} > \frac{\rho}{B} + 1.3\,\frac{\phi}{R'K}. \tag{2}$$

Payload size cancels, so size enters as separate rules: at least two nodes and 32 MiB, and pandas work below 8 FLOP/B (grouping $\phi = 4$, joining $\phi = 2$, $R = 1$ GFLOP/s) stays local up to 512 MiB. A matrix product ($\phi = \max(8, n/8)$, $\rho = 5$, $R = 200$ GFLOP/s) that does not fit in memory moves regardless, or raises an error if it fits nowhere. One GPU peer sets $R' = 6R$ for all K nodes, a coarse heuristic. The test runs first at 10 Gbit/s, and only if offloading could gain is bandwidth measured (Fig. 2). Indivisible work (SVD) moves when its transfer costs under half the local time. Local BLAS outruns a CPU cluster's transport, so dense products stay local.

F. Degradation Guarantee and Operation

Using the cluster must never change the result. Parallel-map parts fill a result table in which the first result for each part is kept, idle local cores re-run any part not yet returned, and the call returns once every item is covered, so an unresponsive cluster degrades to a local pool. Every intercepted call falls back to the original on exception, a failed peer is skipped, a dead worker is respawned, a failed job is rescheduled with back-off, and a training ring continues on the ranks that are left.

A run behaves like a local one. An interrupt sends one cancel frame, which stops the job's process group and its parts on every peer. A dropped connection never kills the job: the daemon logs every output frame and the submitter can reattach and replay it. Each run ends with a line splitting its time between the script and PipedPeer.

G. Implementation

PipedPeer is one Go 1.25 binary, client and daemon, of 46 Go packages and an embedded Python shim; it links peers with quic-go, builds environments with Nix flakes, and runs each job in a `crun` [32] sandbox. A mutation audit disables each of 95 features in turn, and the tests detect every one.

IV. Evaluation

We ask four questions.[1] **Q1:** What does interception cost, including when it declines? **Q2:** How much do the environment cache and the closure diff save? **Q3:** Does work cross machines with correct results, and does it pay between physical machines? **Q4:** Does split training keep accuracy, and when is it faster?

A. Setup

Labs run on an Intel Core i5-13420H laptop (12 threads, 15 GiB, Fedora 43) and an AMD Ryzen 7 5800H laptop (16 threads, about 14 GiB, RTX 3060, Fedora 44). The *plain lab*'s three worker containers share the host's processor and memory: it measures correctness, overhead and caching, not speedup. The *quota lab* gives each container fixed processor and memory limits and shapes links with `tc netem` to LAN, Wi-Fi or 5G; speedup is one container against several of the same size. NAT traversal is tested on a two-site testbed: two containers on separate 192.168.0.0/24 networks, each behind a router container that translates addresses and drops unsolicited packets, the introducer on a third network; every user command ran across it. Two physical sets ran over Tailscale (WireGuard [36]). One joined the Ryzen laptop and a laptop in another home (Core i5-13420H, RTX 3050, 15.3 GiB; 6–8 ms) plus a Galaxy S20 FE phone; neither home offered a reachable introducer, so these runs measure distribution, not traversal. The other joined a 20-core 13th-generation Core i9 machine and the Ryzen laptop, with a public cloud host as introducer. Accuracy is against one device: two ranks in the lab, more in-process. Lab times are medians of 3–5 runs,

[1] Measurement records: `docs/benchmarks` in the project repository [37].

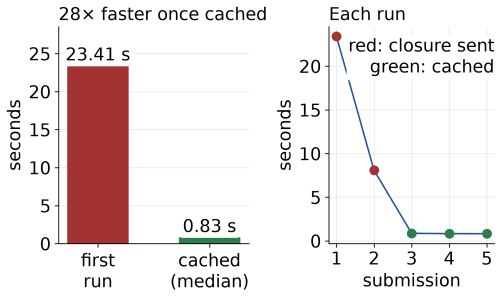


Fig. 3. Five submissions of the same environment in the container lab.

training times one seed, startup the best of five. Accuracy is a mean of three seeds, and in-process also of seeds 0–5 for MLP-C and the transformer. MLP-G and MLP-C share a 559k-parameter perceptron (147 and 180 steps); the transformer has 70k (314 steps). MiB is binary, MB decimal.

B. Microbenchmarks (Q1, Q2)

Interception. On the Core i5 laptop, startup of a job that imports nothing heavy takes 0.117 s with interception against 0.094 s without (1.24×). On a declined job, a 16-item map with 0.01 s of work, the median was 0.80 s against 0.68 s: declining costs about 0.12 s per job.

Environment cache. The first submission of a job took 23.4 s. The second landed on a worker that had not yet received the closure and took 8.1 s; the next three found it in place, shipped only the workspace, and took a median of 0.83 s (Fig. 3). Against a container peer on the same host, adding one package to an environment the peer held sent 28.6 MB instead of 206.0 MB, 86% fewer bytes.

Work split. On a lab limited to 8, 2 and 1 CPUs, a headcount split took 3.6 s, 1.6× slower than local, which puts the local run near 2.3 s; the measured split declined the 1-CPU worker and took 2.9 s, still about 1.3× slower than local. With one peer at a tenth of the CPU, the slowdown against an equal cluster fell from 1.6× on run one to 1.08× by run four as the rate estimates converged.

Introducer. The cloud introducer carried 0.2 MB against the peers' 30 MB; suspended, a known peer was still reached in 0.03 s.

C. End-to-end runs (Q3)

With distribution forced in the plain lab, the node running a 48-item parallel map sent 18 items to three other daemons, one of them the submitter's, with results identical to a local run. In a LAN lab of four containers on the Ryzen laptop, a parallel map took 43.5 s on its first run and 17.0 s with its environment cached, and a GPU job submitted from a node without a GPU ran on the GPU node in 7.6 s cached. An interrupt stopped work on every peer within 2 s. After the introducer changed address, a three-node cluster kept or re-formed its links with no command. A detached job returned its results on reattach, and a two-rank ring finished its 80 epochs in 90.5 s after a rank was killed.

TABLE II
PHYSICAL MACHINES OVER TAILSCALE. SCRIPT TIMES, EXCEPT THE LAST ROW (FROM SUBMISSION); SESSION 2 RAN A LATER BUILD

Measurement	Result
Pool map, Core i9 alone vs. with Ryzen peer	5.8–6.4 vs. 4.9–5.4 s
CPU script, one vs. two laptops	17.4–18.1 vs. 13.3–14.6 s
Training, session 1: 1 GPU / exact / fast / auto (could not switch yet)	4.2 / 37.1 / 7.4 / 32.9 s
validation loss	0.0794 / 0.0797 / 0.0796 / —
Training, session 2: fast / auto	8.5 / 14.6 s
validation loss	0.0797 / 0.0796
Combined script, first run vs. repeat	506 vs. 17.7 s

Table II lists the physical-machine results (first row: second set). A parallel map and a mixed CPU script (pool, joblib, NumPy, pandas) ran faster on two machines than on one, the latter with half its items on the second laptop. The first run of the combined script took 506 s from submission, 487 s of it building its GPU environment (mostly CUDA PyTorch); a repeat took 17.7 s (15.6 s in the script). With a phone added, a laptop's CPU script spread over both laptops and the phone gave the phone 8 of 192 items with an unchanged result, after the phone had built the environment from the cache in 27 s; of 160 small torch models it trained 6, and of 32 none.

Across two NATs. On the two-site testbed, scripts, parallel maps, remote runs and both training sync modes ran across the sites; when a node's link was cut mid-run, its items re-ran on the submitter with the result unchanged.

D. Training (Q4)

When splitting pays. In the lab, a small model with heavy steps split over two containers was faster than one (9.1 against 11.8 s), while a 9 MB model with light steps over Wi-Fi ran ten times slower split than alone (80.7 against 8.0 s). Between the two laptops no split was faster than one GPU (Table II). With the ring fixed by `--ddp 2`, automatic sync synchronised exactly for 19 steps, until its first check at the 20th sync, then switched (25 syncs in all) and finished in 14.6 s, against 8.5 s for fast sync, whose exact warm-up is 15 of the 147 steps; without `--ddp` N the lead releases a ring that one rank alone outruns. Offered 16-, 8- and 2-CPU ranks (containers with CPU quotas on one host), admission took the first two, which ran in 23.6 s with measured batch sizes against 25.4 s with equal ones; one machine beat a three-rank ring there (6.9 vs. 36.4 s).

Fast synchronisation. At the default two-rank interval, plain averaging every 32 steps kept validation loss within 1% of one device in-process: +0.06% on a GPU-sized MLP (MLP-G), +0.64% on a CPU-sized MLP (MLP-C) and +0.73% on a small transformer (seeds 0–2; +0.35% and −1.02% for the last two over seeds 0–5; seeds 3–5 were run for MLP-C and the transformer only). In the lab at 32 steps (one seed), each rank sent 7.6–9.2× fewer bytes than exact sync, and the ring finished 1.2–1.4× faster than exact on LAN and 1.6–3.1× on Wi-Fi. At 64 steps the lab stayed within

the bar too (three seeds, LAN and Wi-Fi, worst $+0.64\%$), but in-process the transformer failed on seeds 0–2 ($+2.13\%$). DiLoCo's outer step left the transformer 21.7% above one device even at one sync per epoch.

Many ranks. In-process runs set $H(n)$ as the largest interval that passed on every model: 32 at two ranks, 16 at three, 4 at four; at eight the transformer failed at every interval down to 2. Of ten methods tested at 4, 8 and 11 ranks, only exact sync with rank-16 PowerSGD [33] and error feedback kept every model within 1% (worst mean $+0.18\%$), with $3.2\times$ fewer bytes than 8-bit values on the MLPs but $1.7\times$ more on the transformer.

V. Limitations

The labs share one processor among their workers, so they show cost, not speedup, and the physical runs cover two or three machines with few repetitions; no wall-time comparison with Ray, Dask or `torchrun` was run, so Table I is qualitative. TLS is trust on first use, the keys trusted to sign closures are not yet bound to an authenticated identity, and authorisation is one shared secret. Only Python is supported, and a training ring depends on rank 0's machine. Fast sync was never measured at five to seven ranks. The 1% accuracy bar is a mean over seeds; single seeds spread from -3.9% (lab, 64 steps) to $+2.2\%$ (in-process, 32).

VI. Conclusion and Future Work

PipedPeer runs unmodified Python on other machines without configuring a cluster, shipping environments as closures diffed by store path, substituting the script's parallel primitives, dividing work by measured speed, and joining networks over direct authenticated links with no relay or privileged node. Its central rule is to measure before distributing and decline when slower. Future work is to bind signing keys to authenticated identities and require TLS, to add locality- and topology-aware placement and AMD accelerators, to let a training ring survive its submitter and rank 0, and to run a controlled evaluation on more machines.

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